What's in a prompt? Language models encode literary style in prompt embeddings

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Abstract

Large language models use high-dimensional latent spaces to encode and process textual information. Much work has investigated how the conceptual content of words translates into geometrical relationships between their vector representations. Fewer studies analyze how the *cumulative* information of an entire prompt becomes condensed into individual embeddings under the action of transformer layers. We use literary pieces to show that information about intangible, rather than factual, aspects of the prompt are contained in deep representations. We observe that short excerpts ($10 - 100$ tokens) from different novels separate in the latent space independently from what next-token prediction they converge towards. Ensembles from books from the same authors are much more entangled than across authors, suggesting that embeddings encode stylistic features. This *geometry of style* may have applications for authorship attribution and literary analysis, but most importantly reveals the sophistication of information processing and compression accomplished by language models.

1 Introduction

"What's in a name?" famously asked Juliet (Shakespeare, ca. 1599) to interrogate the relationship between a concept's multifaceted reality and its shorthand designation in the form of a word.[1]

Four hundred years later, the question finds renewed significance in the context of large language models (LLMs) (Brown et al., 2020; Touvron et al., 2023; Grattafiori et al., 2024), where words are represented as vectors in a high-dimensional latent space (Mikolov et al., 2013a,b). Much research has attempted to elucidate what information these representations, also called 'embeddings', convey, and how this information is encoded. Some fascinating insights have been uncovered in terms of geometrical relationships between concepts (Mikolov et al., 2013a; Park et al., 2024, 2025).

Yet, word-to-vec(tor) embedding is only the first step. For LLMs, an embedding leaves its starting point and is transported, transformer layer after transformer layer, to a new location that will determine next-token prediction. In the process, it loses its original identity and starts accumulating information about all preceding tokens – and the emergent meaning of their sequence (Fig. 1).

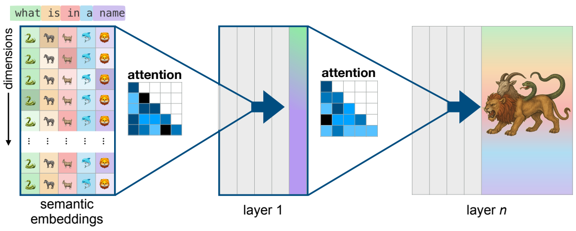

Figure 1: After semantic embedding of the prompt, vectors represent a single word. As the prompt passes through transformer layers, the attention mechanism funnels more and more information about preceding tokens into the last embedding – turning it into a 'chimera' vector, encoding bits of information from all others.

This raises the question: What's in a *prompt*? In other words, what kind of information contained in the sequence of words forming the LLM's input finds itself distilled into deeper embeddings?

Prior work (see also Appendix A) has shown that embeddings can contain global factual information about, e.g., whether the preceding statement is true or false (Marks and Tegmark, 2024), or its relation to space and time (Gurnee and Tegmark, 2024). This is interesting and sensible: factual understanding seems necessary to output compelling prompt continuation.

Here, we find evidence of the presence of more

[1] *"That which we call a rose / By any other word would smell as sweet." Act II, Scene II*

subtle signals. Using short excerpts from various literary works, we show that the embeddings contain implicit information about the origin of the passage and can be classified with high accuracy.

This study is *not* aimed at merely assessing the performance of LLMs for authorship attribution (Huang et al., 2025), but rather at showing that implicit prompt features like authorship are encoded in deep embeddings (and not early ones).

2 Methods

Overview. We base our analysis on ensembles of short excerpts from various literary works and collect the embeddings *of the last (rightmost) token* after each layer of an LLM. From these vector representations, we apply classifier techniques to evaluate whether excerpts can be linked to their original oeuvres based on a *single*, information-rich embedding. We investigate in particular the influence of context length N (number of tokens in the input passage) and layer depth L (number of transformer layers that the prompt has crossed).

Sources. We use digital versions of literary works obtained from the Project Gutenberg website (`gutenberg.org`). We curate a corpus of 19^{th} and early 20^{th} century anglophone novels for both consistency and diversity of styles, some of them from the same author (Appendix B.1).

Processing. A full novel's text is passed through a model's tokenizer. The sequence of tokens' IDs is then split into chunks of length N tokens, with $N = 8, 16, \ldots, 128$ typically. Importantly, these text chunks do not correspond to any particular syntactic unit and can end with any kind of token (Appendix B.1). The rightmost embeddings $\vec{x}_N(L)$ are collected after each layer L for each chunk.[2]

Models. We use a suite of open-source models from `Hugging Face`. We focus on the 16-layer `Llama-3.2-1B` base model (MetaAI, 2024) for insight and extend to other models in Appendix D.

Classifiers. We use standard supervised classifying techniques to investigate the separability of ensembles of high-dimensional vectors: Support Vector Machine (SVM) probes for binary classification, and Multilayer Perceptron (MLP) probes for multiclass (details in Appendix B.1).

[2]Indeed, the last embedding of the prompt is the one that learns from all preceding tokens thanks to the causal masked attention mechanism.

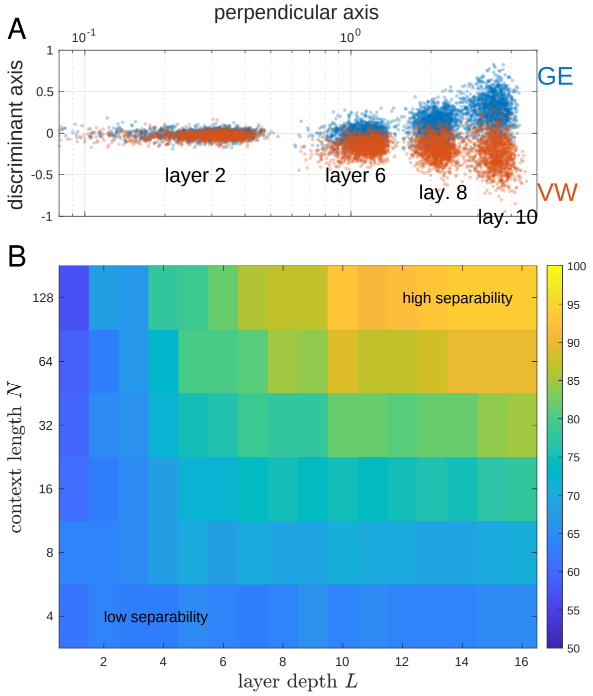

Figure 2: (A) Ensembles of short excerpts ($N = 64$ tokens) from GE and VW separate in the latent space as embeddings travel through successive transformer layers. (B) Linear classifier accuracy (%) to distinguish GE vs VW ensembles as a function of prompt's number of tokens N and number of transformer layers crossed L.

3 Results

3.1 Embeddings encode authorship

Does a short passage (10-100 words) from a novel contain enough information to be properly attributed after processing by an LLM? We compare excerpts from two novels: George Eliot's (GE) *Silas Marner* and Virginia Woolf's (VW) *Mrs Dalloway*. By training a linear classifier, we examine whether the two ensembles of high-dimensional embeddings in the Llama 3.2 1B model can be separated. They can. We observe in Fig. 2 that as their length N increases and the embeddings travel deeper into the model, the excerpts can be classified with over 90% accuracy. In contrast, when there is not enough context (small N) or not enough attention layers to 'cross-pollinate' information across tokens (small L), each excerpt's last embedding has not absorbed enough contextual information to reflect authorship.

More generally, an MLP probe can distinguish excerpts from several novels with overall 75% accuracy (Fig. 3A). These passages represent small snippets of text from various works, with no consistency in theme or syntax (see Tab. 4 in Ap-

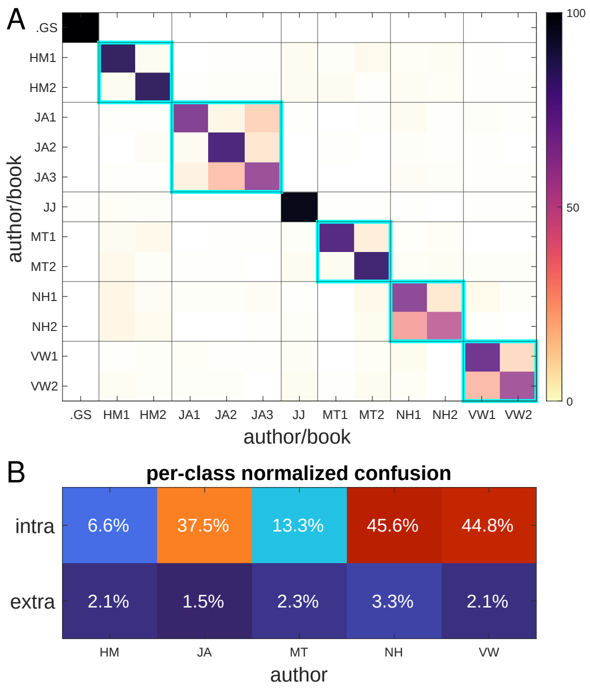

Figure 3: (A) Accuracy (%) of an MLP probe to distinguish passages from 13 different books ($N = 128, L = 16$). See Tab. 3 for the list of authors and novels. Cyan squares emphasize novels from the same authors. It is noteworthy that confusion increases between books of the same author, even though they relate to different topics. (B) Results specific to probe confusion for books from the same author (intra) or a different author (extra).

pendix B.1). It could be that they contain enough *factual* information (names, subjects, etc.) to reveal their provenance. However, we also observe a marked increase in classifier confusion across works from the same authors (Fig. 3B). This suggests that the classifier might be relying on patterns of vocabulary and syntax which find themselves encoded in deep embeddings (and not early ones). We refer to these abstract distinctive features as "style" and investigate what exactly is encoded and how.

3.2 Stylistic signatures align with large principal components

It's been observed that embeddings and their trajectories tend to form low-dimensional structures. For example, Viswanathan et al. (2025) showed that the intrinsic dimension (ID) of token representations from a given prompt is generally much smaller than that of the ambient space. Sarfati et al. (2025) found, using singular value decomposition, that prompt ensembles stretch along a few directions and diffuse about most of the remaining subspace.

Is the property of "style" contained in the small or large dimensions? Fig. 4 indicates that probe accuracy plateaus at a maximum when keeping about 16 directions along the largest principal components. Interestingly, however, the ID of the ensembles remains under 20 dimensions and doesn't change with increased context N. This suggests that contextual information effectively moves ensembles into different corners of the latent space rather than altering their shape complexity.

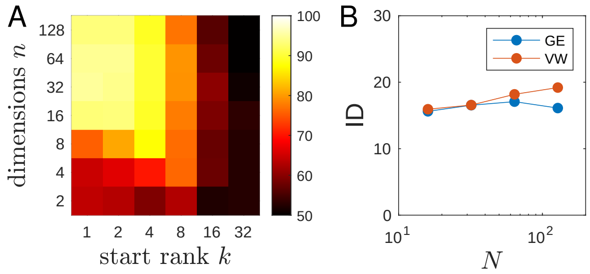

Figure 4: Dimensionality of stylistic features. (A) Probe accuracy (%) for classifying GE vs. VW ensembles projected onto PCA subspaces spanned by $\{\vec{u}_k, \ldots, \vec{u}_{k+n-1}\}$, where $\vec{u}_k$ is the k-th principal component and n is the subspace dimension (B) Intrinsic dimension for embedding ensembles as a function of context length N. ID is calculated using the TwoNN method described in Valeriani et al. (2023).

3.3 Disrupting syntax conserves separability

Is style inferred from syntactic or semantic features? To investigate, we use a shuffling approach introduced in Viswanathan et al. (2025). For each input pseudo-sentence, blocks of B consecutive tokens are rearranged randomly, with $B = 1$ (every token is independent), $B = 4$ (groups of four tokens are kept together), etc. Perhaps surprising, Tab. 1 shows that classifier probes remain accurate for all persistence lengths B. This strongly suggests that the stylistic signature perceived by LLMs might rely more on lexical content than structure. However, ensembles corresponding to different shuffling scales are perfectly separable, indicating that word order and syntax do affect the representation of a prompt – as expected.

VW \ GE	$B = 1$	$B = 4$	$B = 32$
$B = 1$	97	100	100
$B = 4$	100	95	98
$B = 32$	100	99	92

Table 1: Linear probe accuracy (%) for various shuffling block size B.